

Digestive system

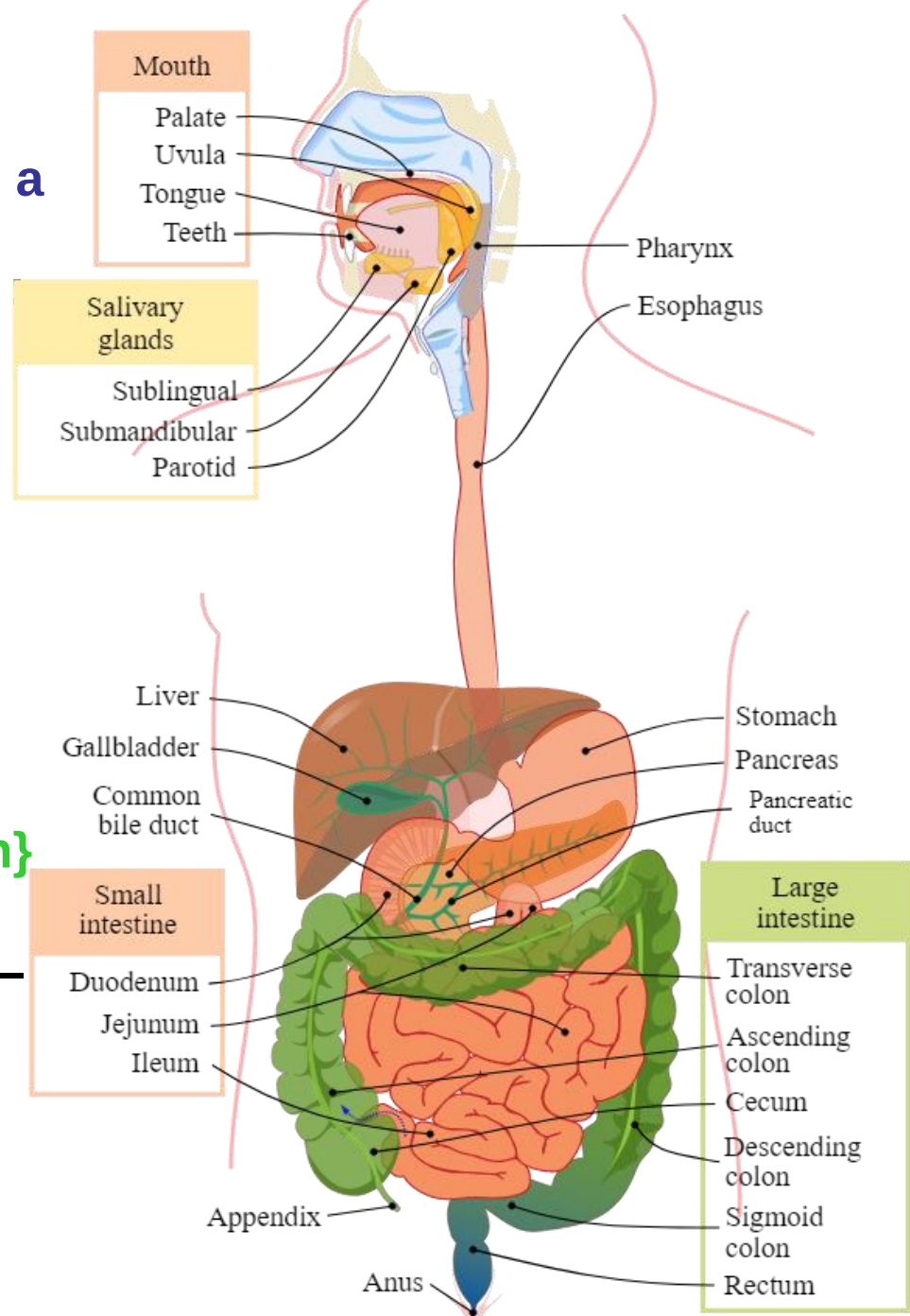
**The basic layers of the wall of
gastrointestinal tract**

LG1

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- **The digestive tract :**
- **is a series of organs forming a long muscular tube, the organs include the following**
- **Oral cavity**
- **pharynx**
- **Esophagus**
- **Stomach**
- **Small intestine (duodenum, jejunum & ileum)**
- **Large intestine (cecum, appendix, {ascending-transverse-descending colon}, rectum & anal canal)**
- **Accessory digestive organs – teeth, tongue, gallbladder, salivary glands, liver, and pancreas**



Histology of the Alimentary Canal

- From esophagus to the anal canal the walls of the GI tract have the same four tunics:

1-Mucosa

a-epithelial lining

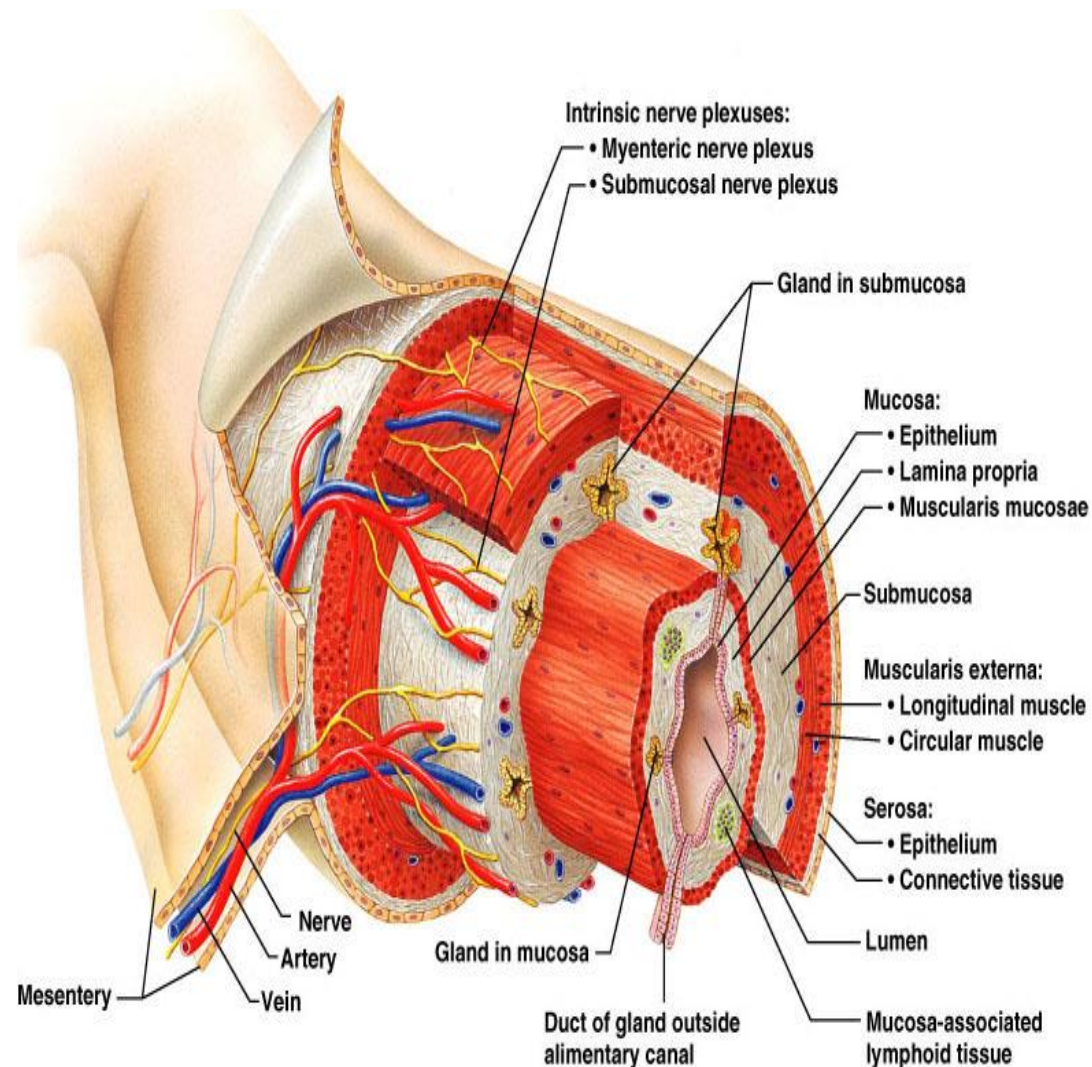
b-lamina propria

c-muscularis mucosa

2-Submucosa

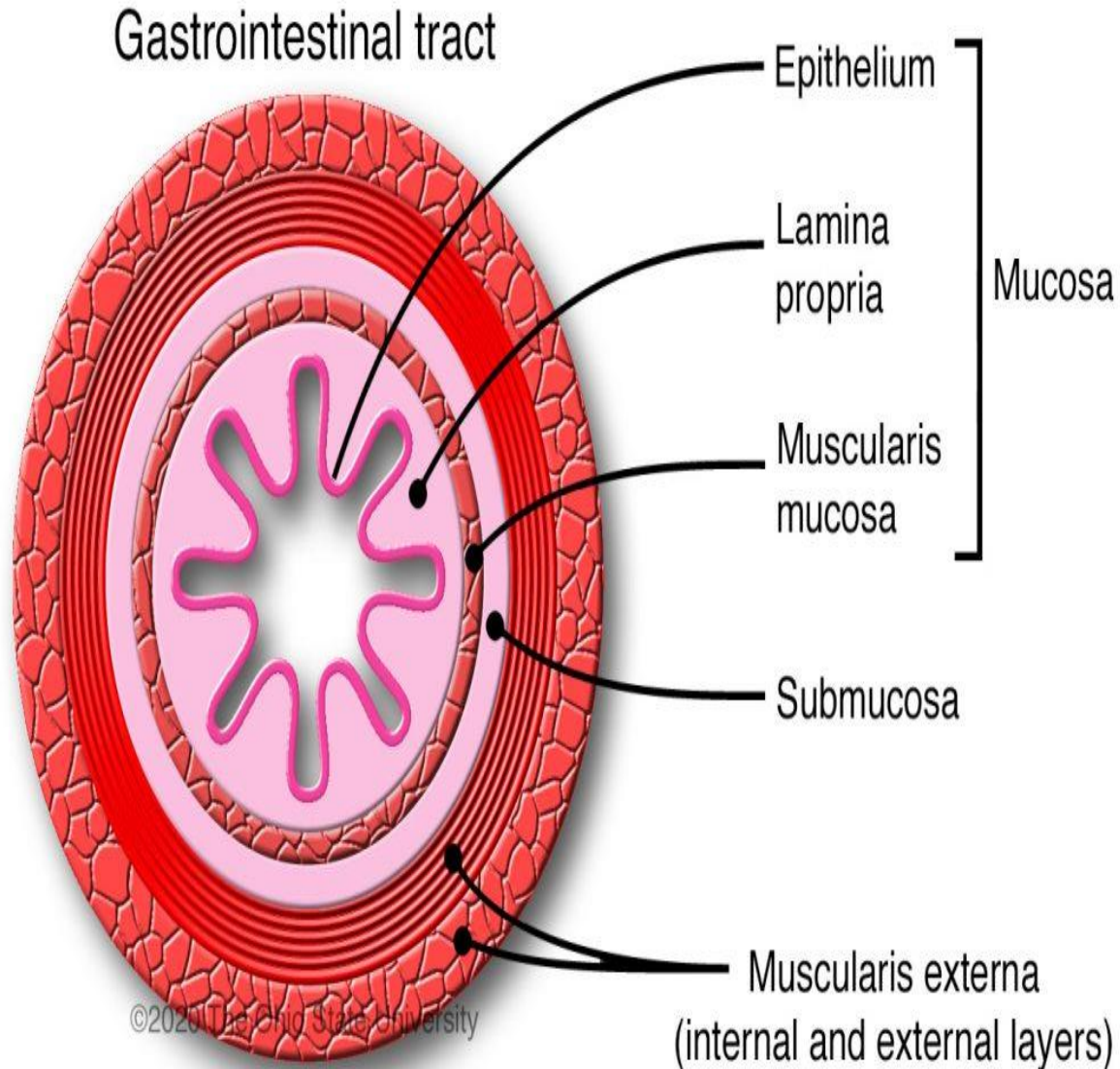
3-Muscularis externa

4-Serosa, or adventitia



Mucosa

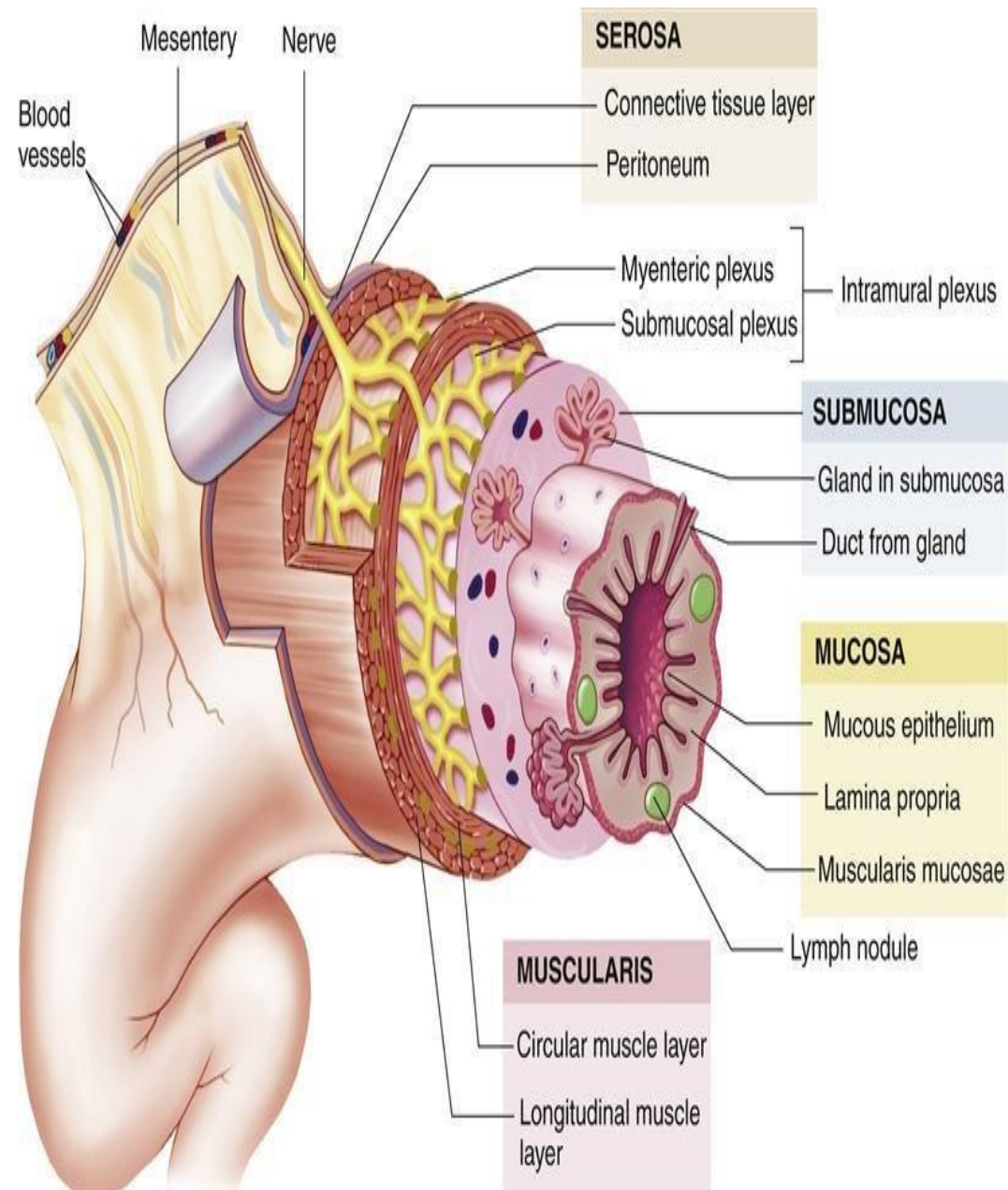
- **The main functions of the mucosa**
 - Secretion**
 - absorption**
 - Protection**
- **Consists of three layers:**
 - **a lining epithelium**
 - **lamina propria**
 - **muscularis mucosae**



Mucosa

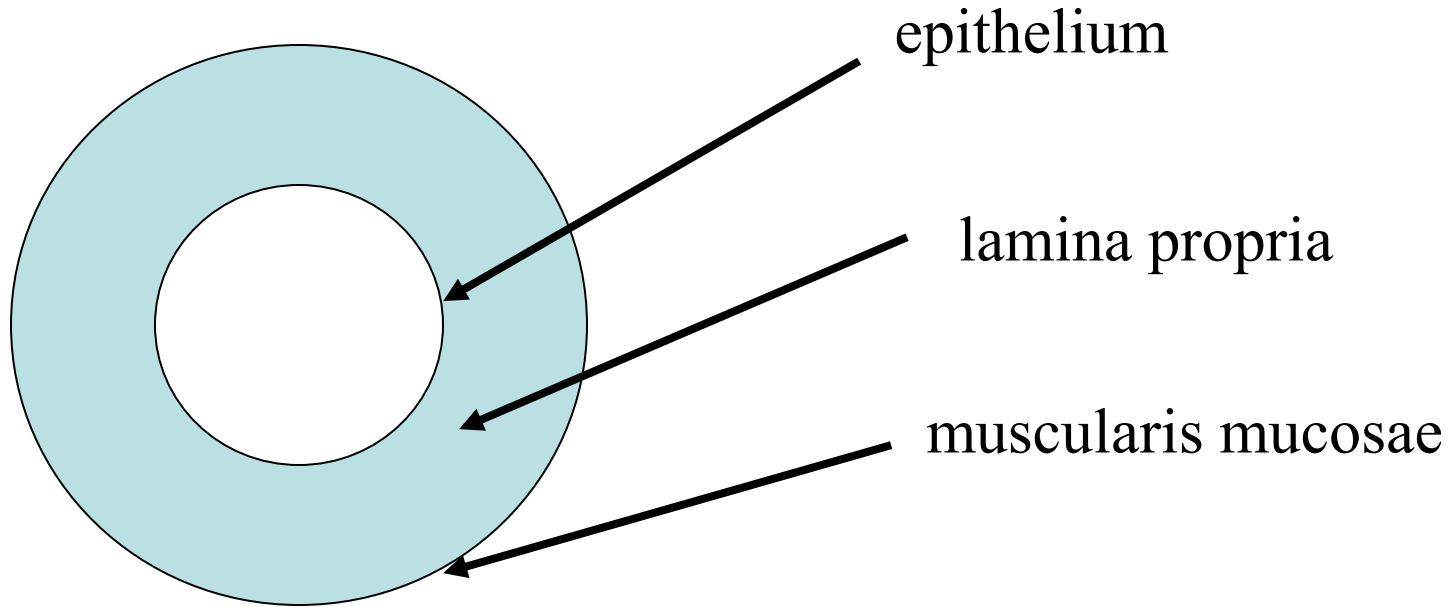
- **Epithelial** The epithelial lining of the mucosa varies from stratified squamous that able to resist abrasion, to tall columnar with a brush border, suitable for absorption
- **Lamina Propria** : The lamina propria is a loose connective tissue layer that supports the epithelium, contain blood vessels & lymphatic tissue , also contain glands open onto the epithelium.

Muscularis mucosae – is an incomplete layer of smooth muscle cells between mucosa & submucosa.



Characteristics of the GI Tube

Mucosa



Epithelium: stratified squamous --> simple columnar --> stratified squamous

esophagus

stomach --> rectum

anus

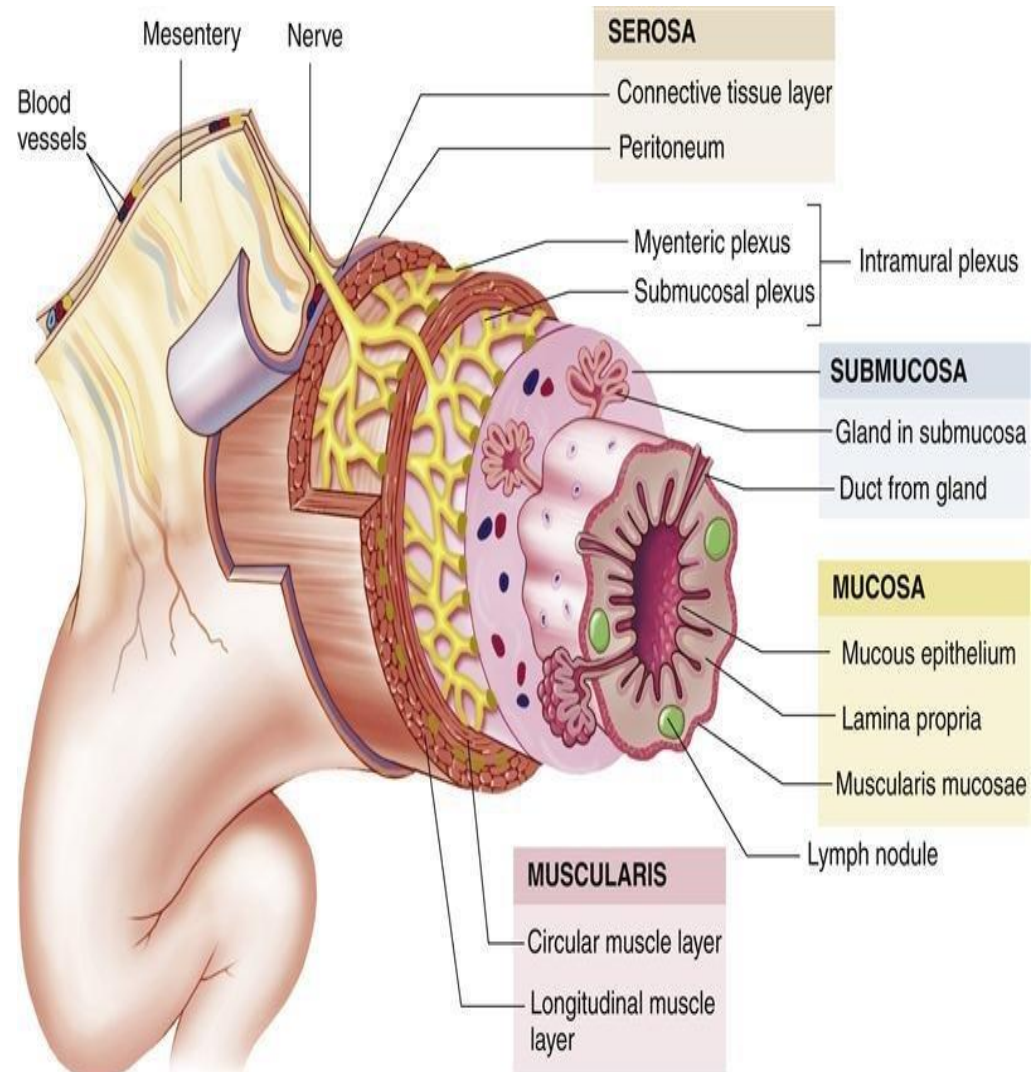
Lamina propria: loose CT - rich in cells, capillaries, lymphatic vessels

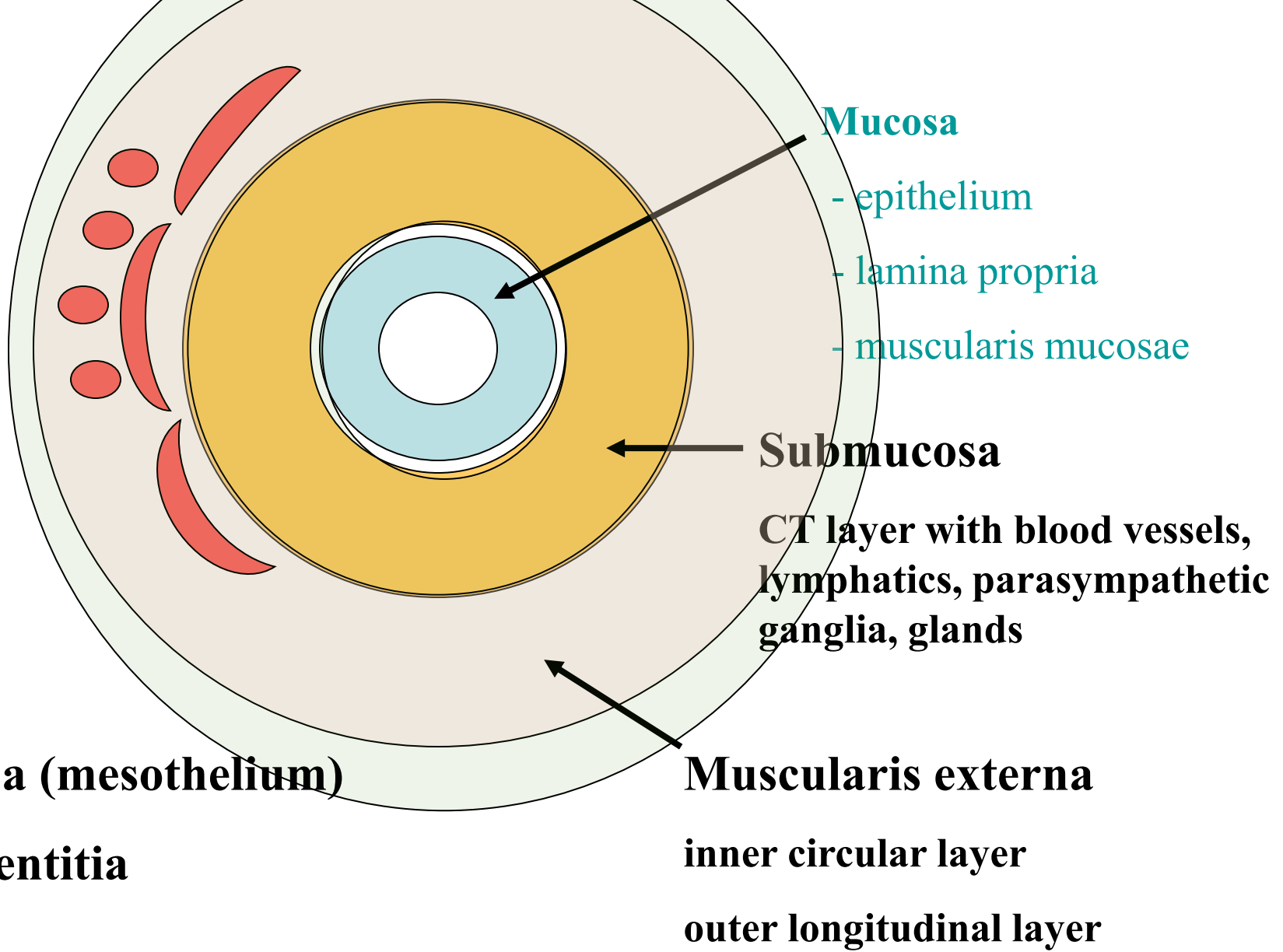
Muscularis mucosae: smooth muscle, two layers

- **Submucosa** – dense connective tissue containing blood and lymphatic vessels. With Meissner's plexus.

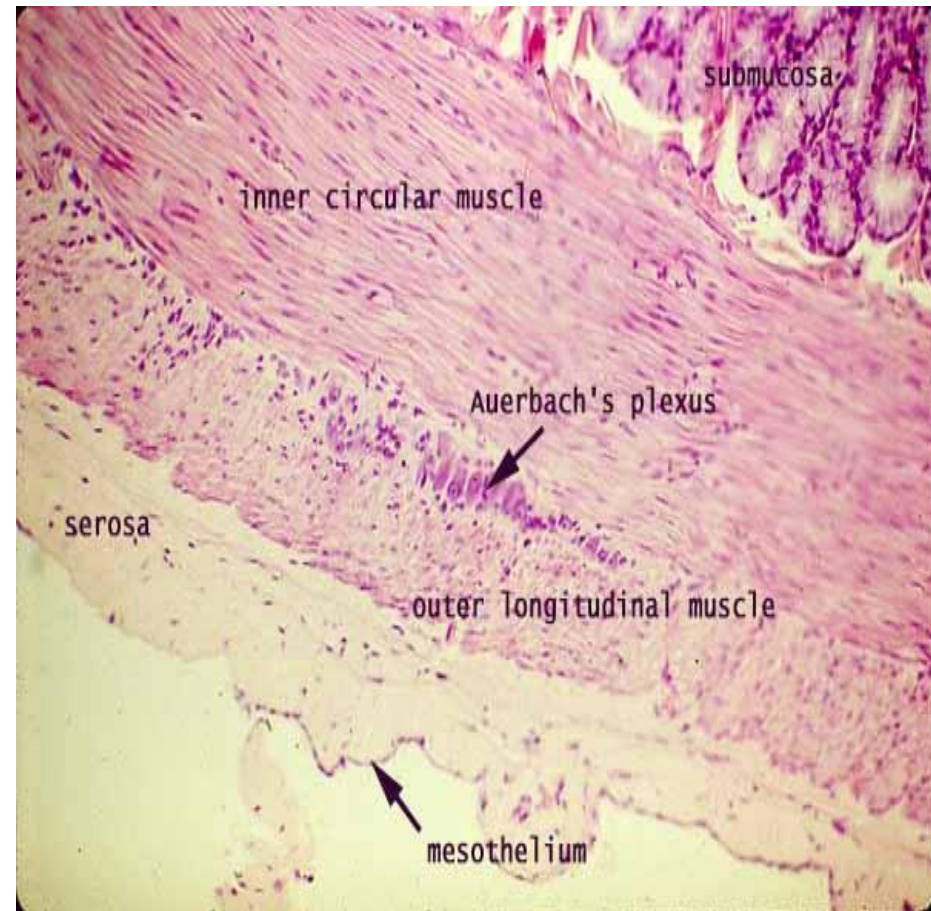
Muscularis externa – is the muscle coat of the gut tube. it is usually formed of an inner circular & outer longitudinal layers of smooth muscle.

• In some parts striated muscle replaces the smooth muscle as in esophagus & in other 3 layers of smooth muscle may be present as in stomach. Auerbach's plexus (Myenteric plexus) found between the muscle coat & controls their function.





- **Serosa** – is the outermost layer of the gastrointestinal tract wall for organs located inside the peritoneal cavity
- **Structure and Composition**
- **Mesothelium:** A simple squamous epithelium covering the outer surface and connective tissue a thin layer of loose connective tissue beneath the epithelium
- **Location** :Intraperitoneal Organs, such as the stomach, jejunum, ileum, and transverse colon

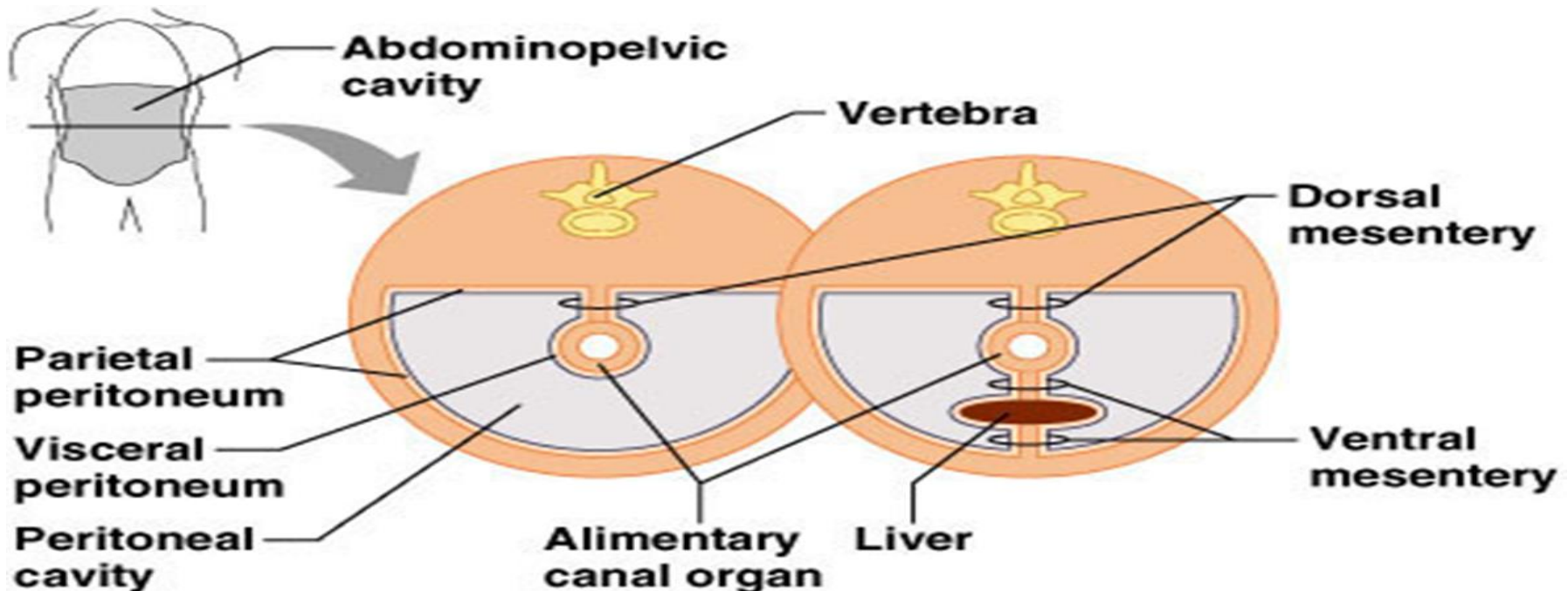


replaced by Adventitia in parts fixed outside the cavity, like the thoracic esophagus, most of the duodenum, and the ascending/ descending colon

Functions: Lubricates organs so they move freely during digestion
 .Protection: Acts as a smooth, tough barrier against physical wear.
 Vascular Pathway: Transmits blood vessels, nerves, and lymphatics to the wall

Peritoneum and Peritoneal Cavity

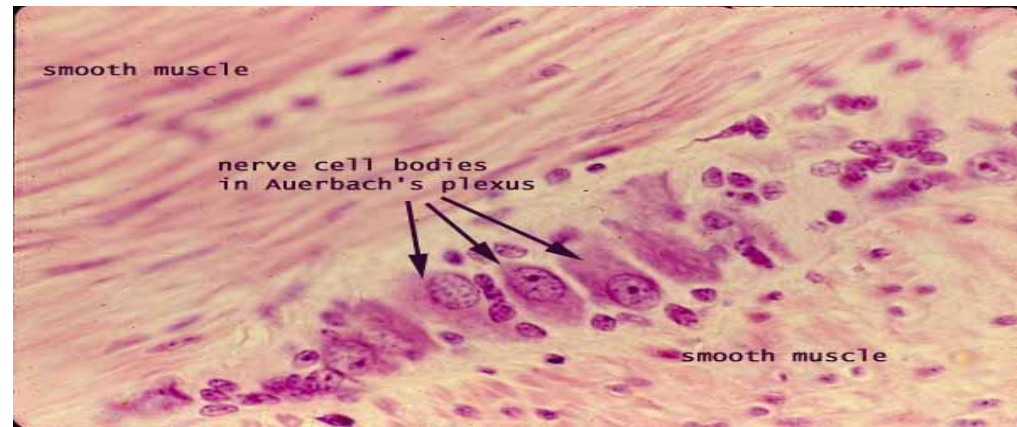
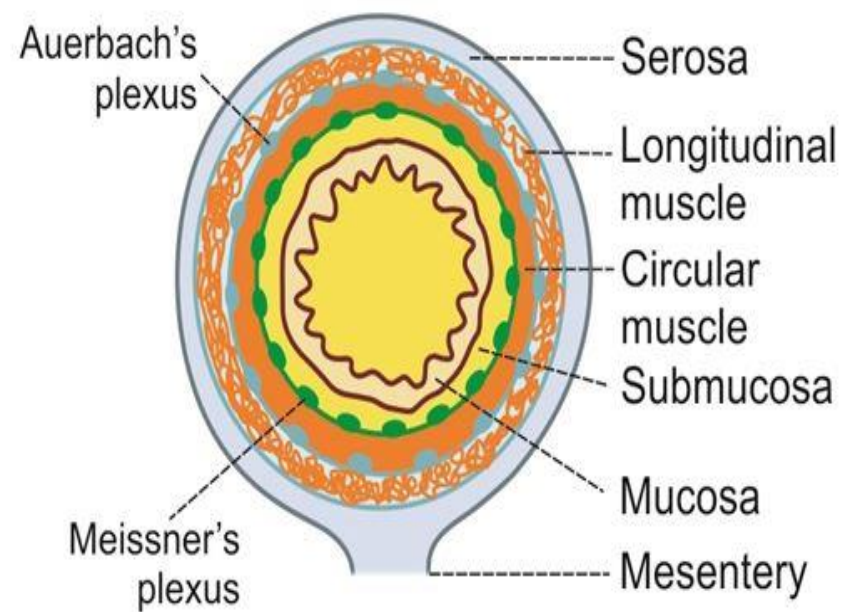
- **Peritoneum** – serous membrane of the abdominal cavity
 - **Visceral** – covers external surface of most digestive organs
 - **Parietal** – lines the body wall
- **Peritoneal cavity**
 - Lubricates digestive organs
 - Allows them to slide across one another



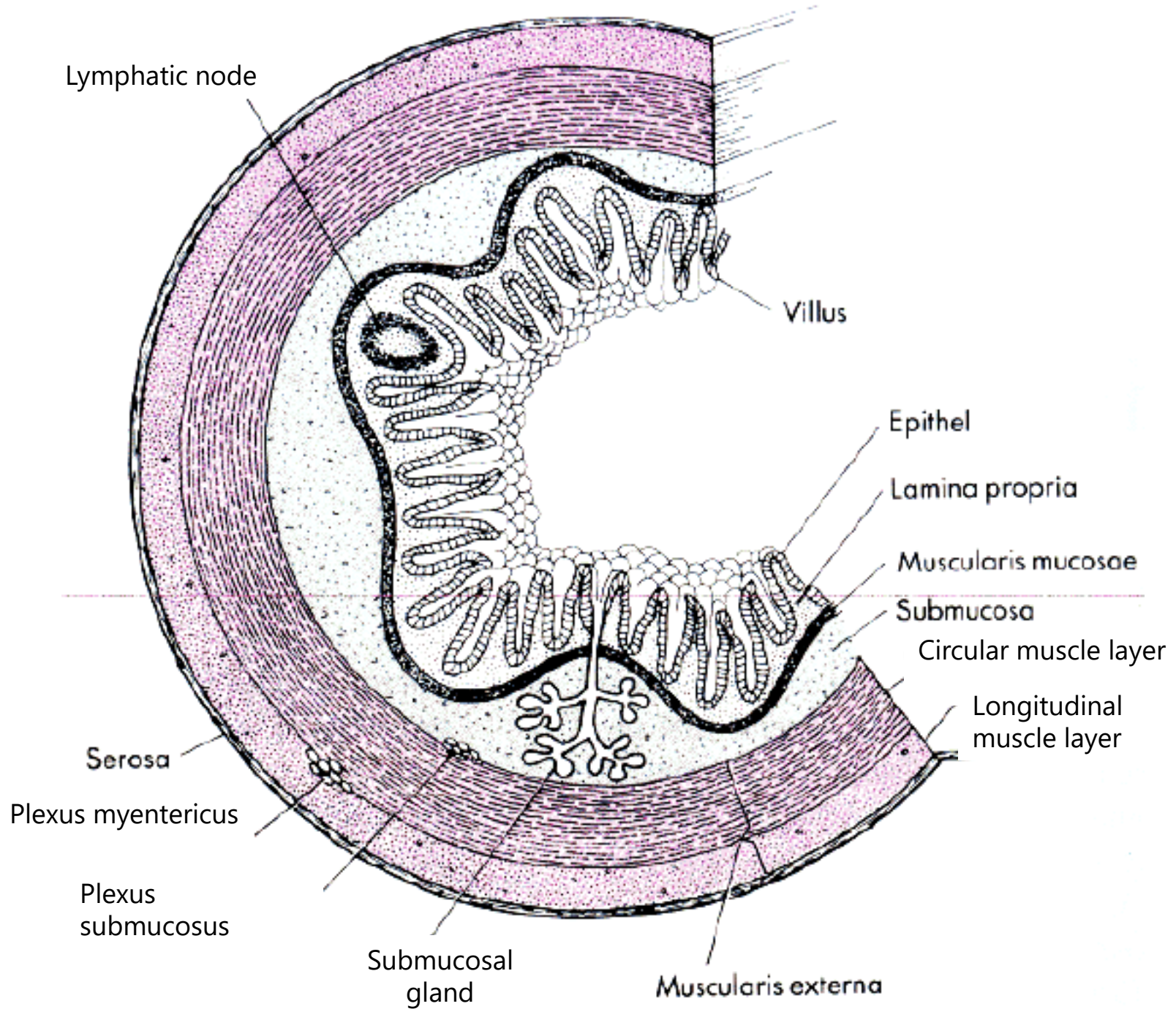
(a) Transverse section of abdominal cavity

Enteric Nervous System

- These are Ganglia(Clusters of Neurons and supporting glial cells). Composed of two major intrinsic nerve plexuses
- **Submucosal plexus (Meissner's plexus):** Located inside the submucosa layer of the gastrointestinal wall. It regulates local blood flow, gut absorption, and the secretion of digestive fluids and enzymes.

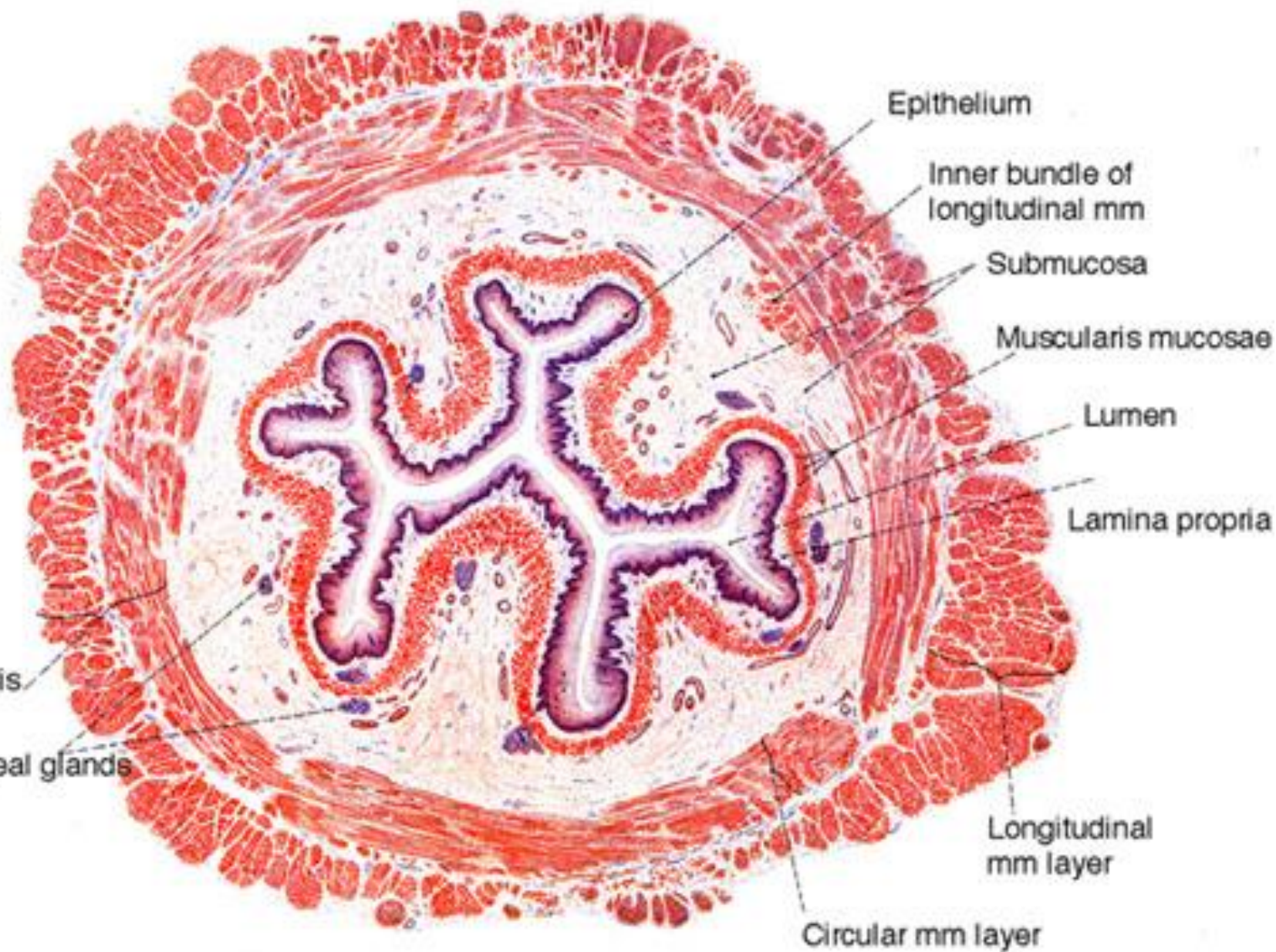


Myenteric nerve plexus Myenteric plexus (Auerbach plexus): Located between the inner circular and outer longitudinal muscle layers of the muscularis externa, controls muscle contractions, movement, and peristalsis.

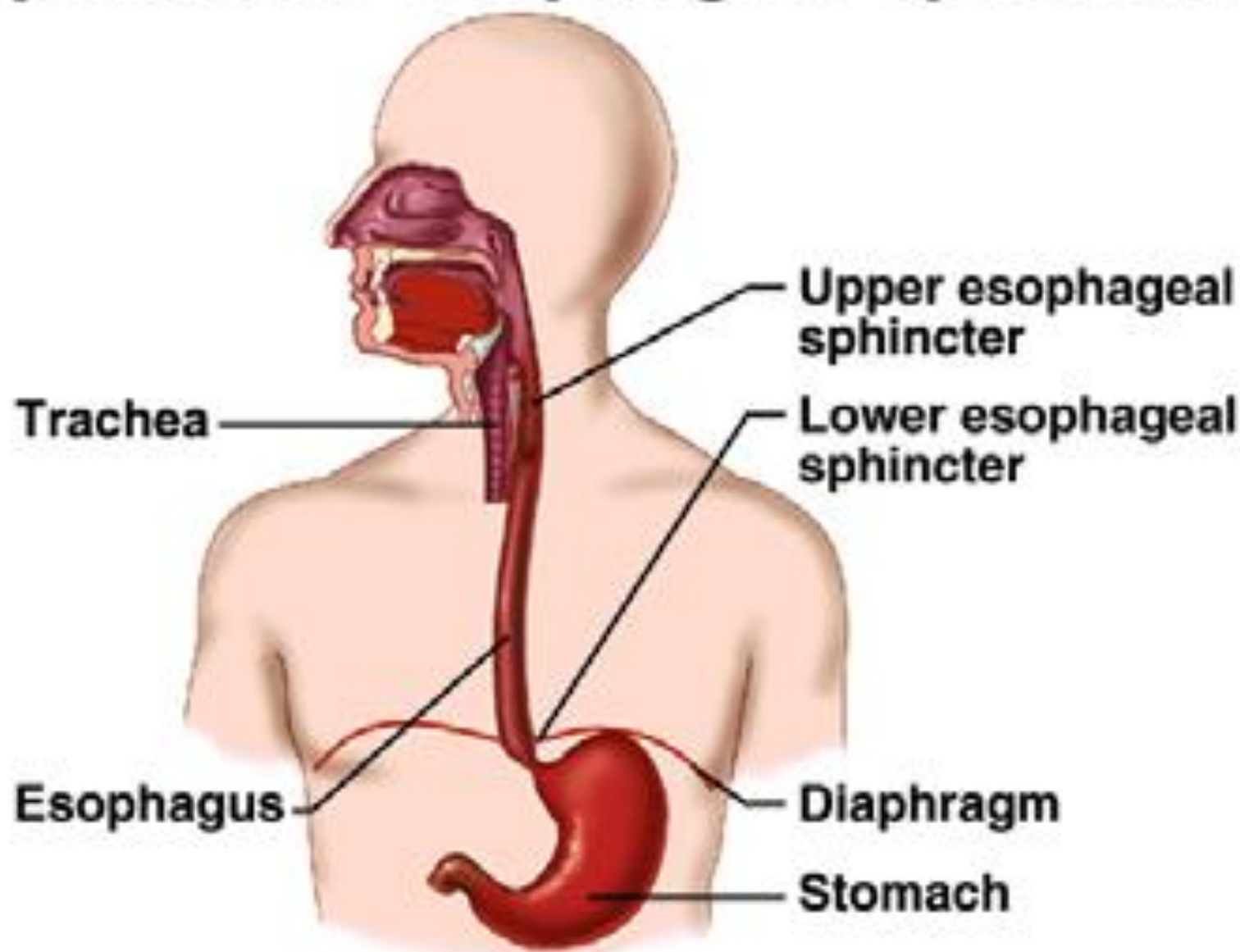


Esophagus

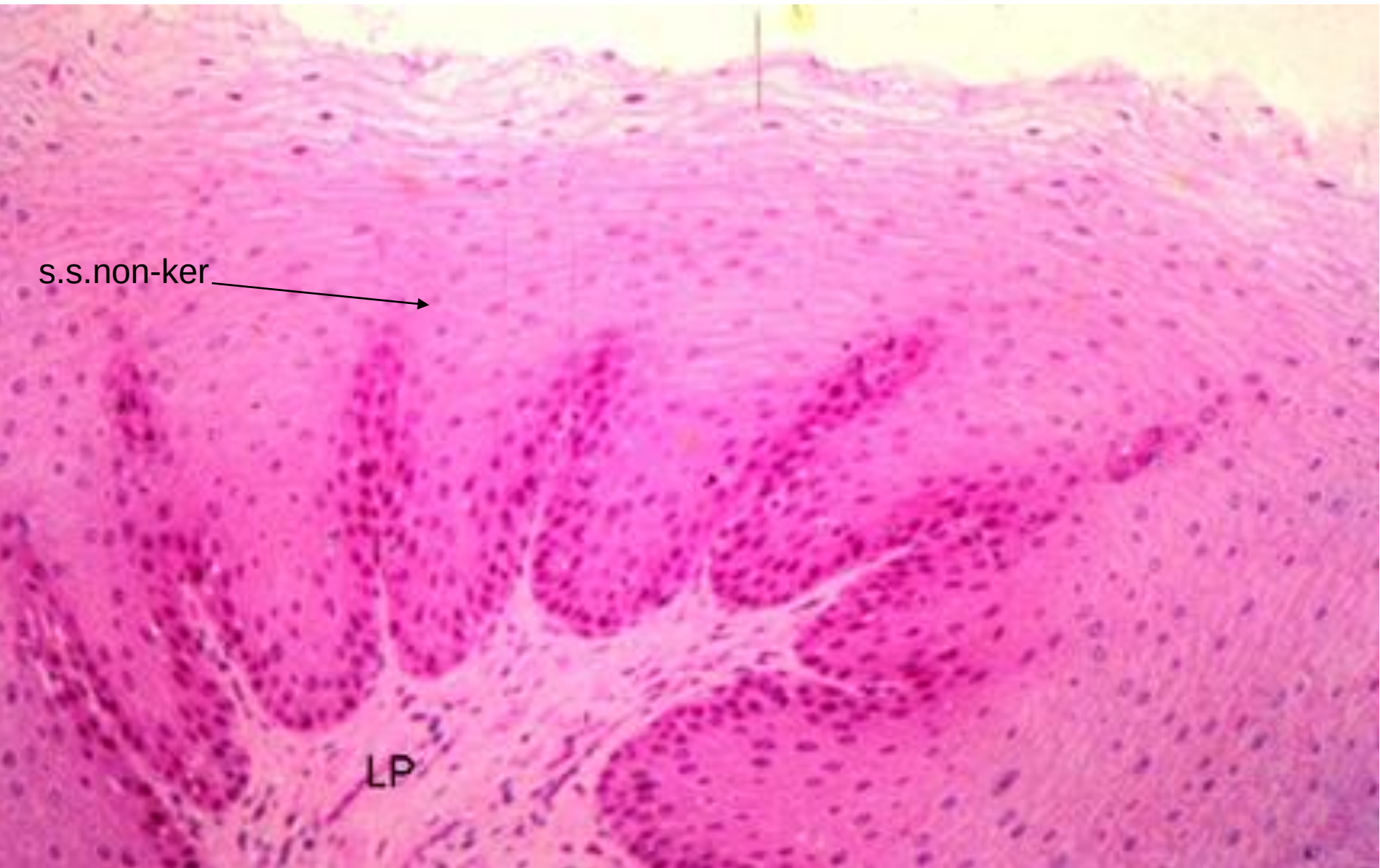
- The esophagus extends from the pharynx to the stomach. Most of the esophagus lies in the thorax and only the last 2-3 cm are within the abdominal cavity . Their wall consist of 4 parts:
 1. **Mucosa**: lined by s.s.ep.non- ker., the lamina propria consist of loose c.t.contain lymphatic nodules , has muscularis mucosa
 2. **Sub-mucosa** : is dense c.t contain mucus g. secrete mucus as a food moves through the esophagus , bundle of elastic & collagen fibers make lumen as stellate shape
 - 3-**Muscularis externa**: is composed of :
 - skeletal muscle** in upper one third ,
 - mixture of skeletal& smooth** in middle one third
 - smooth muscle** in lower one third (**inner circular & outer longitudinal**)
 - 4-**Serosa& adventitia**: only that part of esophagus that is in the peritoneal cavity is covered by **serosa** , the rest is covered by a layer of loose **c.t. the adventitia**



Upper/lower esophageal sphincters



Esophagus



Esophago gastric junction:

is short about (1-1.5 cm.) long of esophagus below the diaphragm , the epithelium change from s. to columnar ep.

in the lamina propria contain group of mucous glands the esophageal cardiac gland that secret mucous

Esophago gastric junction



Thanks